YOUR INSTITUTION NAME

EXAMINATION 2026-27

Subject: Mathematics ریاضی | Class: 10th

PAPER CODE 3 0 4 4

Student Name:		**Syllabus:**	

Roll No: _____ | **Time:** 2:30 hrs | **Marks:** 25 | **Date:** 11-10-2026 | **Test No.:** _____

Note: Four possible answer a,b,c and d to each question are given. The choice which you think is correct, fill that circle in front of that questions with Marker or Pen ink, Cutting or filling two or more circles will result in zero mark in that question.

Section-I

Tick the correct Answer. (Marks:1x10=10)

01. (A) (B) (C) (D)	02. (A) (B) (C) (D)	03. (A) (B) (C) (D)	04. (A) (B) (C) (D)	05. (A) (B) (C) (D)
06. (A) (B) (C) (D)	07. (A) (B) (C) (D)	08. (A) (B) (C) (D)	09. (A) (B) (C) (D)	10. (A) (B) (C) (D)

1. A triangle is isosceles if at least ____ sides have equal magnitude.

(A) Zero (کوئی نہیں) **(B)** Three (تین) **(C)** One (ایک) **(D)** Two (دو)

2. If initial velocity is $\underline{v}$ and final velocity is $\underline{v'}$, the change in velocity is:

(A) $\underline{v} + \underline{v'}$ **(B)** $\underline{v'} - \underline{v}$ **(C)** $\underline{v} - \underline{v'}$ **(D)** $\underline{v} \cdot \underline{v'}$

3. An aeroplane with airspeed $20\underline{i}$ in a crosswind of $50\underline{j}$ has resultant velocity:

(A) $70\underline{i}$ **(B)** $30\underline{j}$ **(C)** $20\underline{i} + 50\underline{j}$ **(D)** $1000\underline{ij}$

4. If vector $\vec{AB} = 6\underline{i} + 2\underline{j}$, then vector $\vec{BA}$ is:

(A) $6\underline{i} - 2\underline{j}$ **(B)** $-6\underline{i} + 2\underline{j}$ **(C)** $-6\underline{i} - 2\underline{j}$ **(D)** $2\underline{i} + 6\underline{j}$

5. Translating point A(-5, 4) by vector $8\underline{i} - 6\underline{j}$ gives point A' at:

(A) (-13, 10) **(B)** (13, -2) **(C)** (3, 10) **(D)** (3, -2)

6. For a velocity vector $10\underline{i} + 20\underline{j}$, the horizontal component is:

(A) 10 **(B)** 20 **(C)** 30 **(D)** 200

7. A square is shifted by a translation vector. This changes the ____ of the square.

(A) Shape (شکل) **(B)** Position (مقام) **(C)** Size (سائز) **(D)** Area (رقبہ)

8. If a point (x, y) is translated by a vector $a\underline{i} + b\underline{j}$, the new coordinates are:

(A) (x-a, y-b) **(B)** (ax, by) **(C)** (x+a, y+b) **(D)** (x/a, y/b)

9. The magnitude of the translation vector $6\underline{i} - 6\underline{j}$ is:

(A) 12 **(B)** $6\sqrt{2}$ **(C)** 0 **(D)** 6

10. In a parallelogram ABCD, the vector $\vec{AB}$ is equal to:

(A) $\vec{DC}$ **(B)** $\vec{CD}$ **(C)** $\vec{BC}$ **(D)** $\vec{AD}$

Section-II (Marks: 24)

Q. 2: Write the short answer of any 6 parts of the following. (Marks:2x6=12)

(i) Plot A(-5, 4), B(-6, 1) and C(-3, 1) to form a ΔABC. Also translate ΔABC to $\Delta A'B'C'$ by translation vector $8\underline{i} - 6\underline{j}$.

(ii) Plot A(-6, -2), B(-6, -5), C(-3, -5) and D(-3, -2) to form a square ABCD. Also translate square ABCD to square A'B'C'D' by translation vector $9\underline{i} + 7\underline{j}$.

(iii) Plot A(-6, 3), B(-4, 0), C(-2, 3) and D(-4, 4) to form a kite ABCD. Also translate kite ABCD to kite A'B'C'D' by translation vector $6\underline{i} - 6\underline{j}$.

(iv) Use vectors to show that triangle XYZ is an isosceles, where the points X, Y, Z have the coordinates (0, 0), (2, 0) and (1, 3) respectively.

ANSWER KEY

Multiple Choice Questions

Q. No.	Correct Answer	Q. No.	Correct Answer	Q. No.	Correct Answer
01	**(D)** Two (دو)	**05**	**(D)** (3, -2)	**09**	**(B)** $6\sqrt{2}$
02	**(B)** $\underline{v'} - \underline{v}$	**06**	**(A)** 10	**10**	**(A)** $\vec{DC}$
03	**(C)** $20\underline{i} + 50\underline{j}$	**07**	**(B)** Position (مقام)		
04	**(C)** $-6\underline{i} - 2\underline{j}$	**08**	**(C)** (x+a, y+b)		